

Experiment 13

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Slot: B

Advanced Policy Optimization using PPO, TRPO and DDPG

AIM

To implement advanced policy optimization algorithms **PPO, TRPO and DDPG** and compare their Reinforcement Learning performance.

PROCEDURE

1. Set up the Python environment with the required Reinforcement Learning libraries.
2. Import Stable-Baselines3, TensorFlow and Gymnasium.
3. Create suitable environments for the selected algorithms.
4. Configure the PPO algorithm.
5. Configure the TRPO algorithm.
6. Configure the DDPG algorithm.
7. Train each algorithm for the specified number of timesteps.
8. Evaluate the trained agents using their obtained rewards.
9. Record the performance of PPO, TRPO and DDPG.
10. Compare the performance of the three algorithms using a graph.

CODE:

```
!pip -q install stable-baselines3 sb3-contrib gymnasium
```

```
import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt
```

```
from stable_baselines3 import PPO, DDPG
from sb3_contrib import TRPO
```

```
env_ppo = gym.make("CartPole-v1")
env_trpo = gym.make("CartPole-v1")
```

```

env_ddpg = gym.make("Pendulum-v1")

ppo_model = PPO(
    "MlpPolicy",
    env_ppo, verbose=0,
    learning_rate=0.0003
)

trpo_model = TRPO(
    "MlpPolicy",
    env_trpo, verbose=0,
    learning_rate=0.0003
)

ddpg_model = DDPG(
    "MlpPolicy",
    env_ddpg,
    verbose=0,
    learning_rate=0.001
)

# =====
# 3. TRAIN MODELS
# =====

TIMESTEPS = 3000

print("Training PPO...")
ppo_model.learn(total_timesteps=TIMESTEPS)
print("PPO training completed.")

print("\nTraining TRPO...")
trpo_model.learn(total_timesteps=TIMESTEPS)
print("TRPO training completed.")

print("\nTraining DDPG...")
ddpg_model.learn(total_timesteps=TIMESTEPS)
print("DDPG training completed.")

def evaluate_agent(model, env, episodes=10):

    rewards = []

```

```

        for _ in range(epochs):

            obs, info = env.reset()
            total_reward = 0            done =
            False

            while not done:

                action, _ = model.predict(
obs,
                deterministic=True
                )

                obs, reward, terminated, truncated, info = env.step(
action
                )

                done = terminated or truncated
            total_reward += reward

            rewards.append(total_reward)

        return rewards

# =====
# 5. EVALUATE
# =====

print("\nEvaluating PPO...")
ppo_rewards = evaluate_agent(
ppo_model,    env_ppo
)

print("Evaluating TRPO...")
trpo_rewards = evaluate_agent(
trpo_model,    env_trpo
)

print("Evaluating DDPG...")
ddpg_rewards = evaluate_agent(
ddpg_model,    env_ddpg
)

```

```

# =====
# 6. DISPLAY RESULTS
# =====

print("\n=====") print("PERFORMANCE
SUMMARY")
print("=====")

print(
    "PPO Average Reward :",
    round(np.mean(ppo_rewards), 2)
)

print(
    "TRPO Average Reward :",
    round(np.mean(trpo_rewards), 2)
)

print(
    "DDPG Average Reward :",
    round(np.mean(ddpg_rewards), 2)
)

# =====
# 7. COMPARISON GRAPH
# =====

methods = [
    "PPO",
    "TRPO",
    "DDPG"
]

average_rewards = [
    np.mean(ppo_rewards),
    np.mean(trpo_rewards),
    np.mean(ddpg_rewards)
]

plt.figure(figsize=(9, 5))

bars = plt.bar(
    methods,
    average_rewards

```

```

)

plt.xlabel("Algorithm")
plt.ylabel("Average Evaluation Reward")

plt.title(
    "Performance Comparison of PPO, TRPO and DDPG"
)

plt.grid(
    axis="y",
    linestyle="--",
    alpha=0.6
)

# Display values above bars
for bar, value in zip(
    bars, average_rewards ):
    plt.text(
        bar.get_x() +
        bar.get_width() / 2,
        bar.get_height(),
        f"{value:.2f}",
        ha="center",
        va="bottom"
    )

plt.tight_layout()
plt.show()

# =====
# 8. CLOSE ENVIRONMENTS
# =====

env_ppo.close() env_trpo.close()
env_ddpg.close()

print("\nExperiment 13 completed successfully!")

```

OUTPUT:

PPO training completed.

Training TRPO...

TRPO training completed.

Training DDPG...

DDPG training completed.

Evaluating PPO...

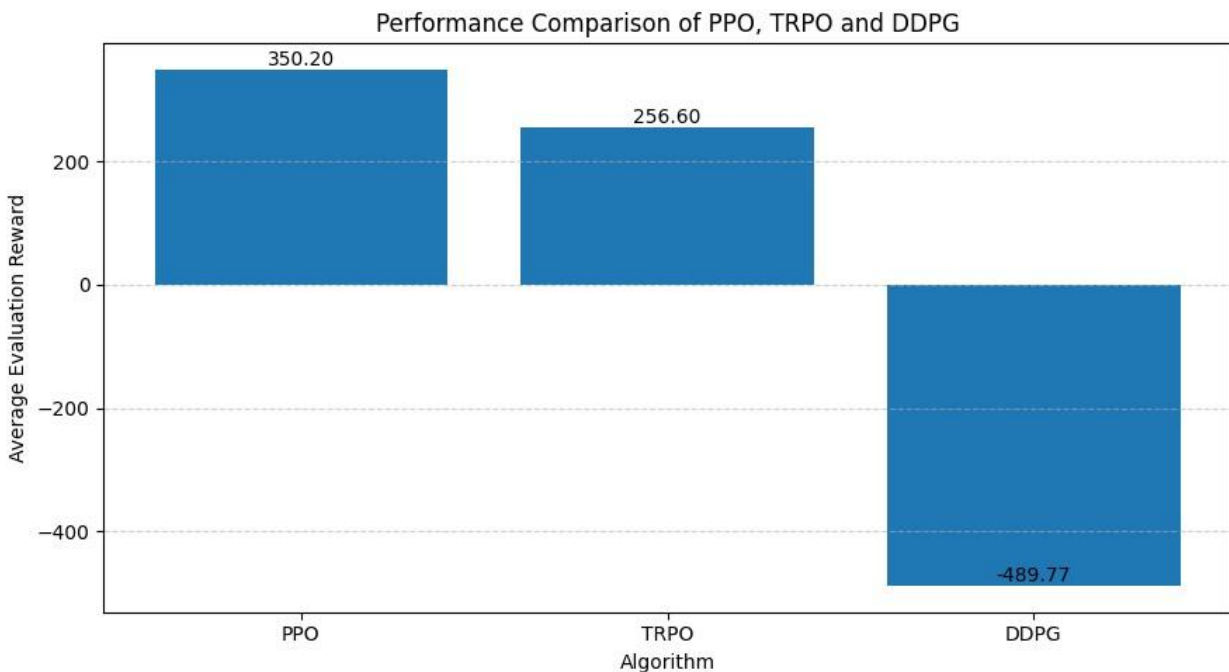
Evaluating TRPO...

Evaluating DDPG...

===== PERFORMANCE
SUMMARY

=====

PPO Average Reward	: 350.2
TRPO Average Reward	: 256.6
DDPG Average Reward	: -489.77



Experiment 13 completed successfully!

RESULT:

PPO, TRPO and DDPG were successfully implemented and their policy optimization performance was compared.